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Final Report

Stock Price Time Series Analysis:

A Comparative Study of Deep Learning Across Market Regimes with Automated Smart Prediction

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Abstract

This report presents a comprehensive comparative study of seven time-series forecasting architectures applied to stock index prediction across seven historically significant market regimes. Building upon an interim study establishing ARIMA, LSTM, and GRU baselines on individual stocks, this final work substantially expands the experimental scope to include Bidirectional RNN (Bi-RNN), Bidirectional GRU (Bi-GRU), Bidirectional LSTM (Bi-LSTM), PatchTST (a patch-based Transformer), Mamba (a selective state-space model), and Spiking Neural Network (SNN), evaluated on the S&P 500 (SP500) and Hang Seng Index (HSI) across prediction horizons of 1, 5, and 10 days and lookback windows of 30, 60, 90, 120, and 252 trading days.

The core contribution is the *Special Period Analysis* framework, which employs a strict walk-forward training protocol to evaluate model performance across seven distinct market regimes: the 2008 Financial Crisis, 2010 Euro Debt Crisis, 2015 China Stock Market Crash, 2018 US-China Trade War, 2020 COVID-19 Crash, 2022 Federal Reserve Rate Hike cycle, and the 2023 AI-driven Bull Market. Additionally, a *Smart Prediction* module is implemented that automatically identifies the statistical characteristics of any input date range, matches it to the most similar historical regime via z-score normalised Euclidean distance, and deploys a market-specific, horizon-aware weighted ensemble.

Key findings: (1) Mamba is the best deep learning model for SP500 short-horizon prediction; (2) the HSI is substantially more predictable than SP500 (overall R^2 of 0.71 vs. 0.38), with deep learning models achieving near-parity with ARIMA on HSI; (3) forecast horizon is the single most influential factor, with R^2 declining 83% from H=1 to H=10 on SP500; and (4) the 2020 COVID crash demonstrates that black swan events remain fundamentally unpredictable for all tested architectures.

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1. Introduction

1.1 Background

Stock market prediction remains one of the most challenging problems in quantitative finance and machine learning. The inherent complexity of financial time series makes accurate forecasting exceptionally difficult. The potential economic value of even marginal predictive improvements has driven substantial research interest from classical statistical methods to modern deep learning architectures.

The Efficient Market Hypothesis (EMH) [6] posits that all publicly available information is instantaneously reflected in asset prices, implying that price movements should approximate a random walk and be inherently unpredictable beyond short horizons. However, a substantial body of empirical evidence has identified exploitable inefficiencies, particularly in markets with higher retail investor participation, regulatory interventions, or structural information asymmetries [17].

A critical gap in the existing comparative literature is the lack of systematic evaluation across *different market regimes*. Most studies evaluate models over a single continuous historical period, obscuring the fact that model suitability varies dramatically depending on whether the market is trending, crisis-driven, or subject to exogenous policy shocks. Trend-following architectures that excel in sustained bull markets may fail catastrophically during sudden reversals, while simpler models may generalise better across diverse regimes. This project addresses this gap through a principled *special period analysis*.

1.2 Problem Statement

This project addresses two central research questions:

RQ1 (Model Suitability by Regime): Do different deep learning and statistical architectures exhibit systematically different performance across historically distinct market periods? If so, what architectural properties explain these differences?

RQ2 (Automated Regime-Aware Prediction): Can the patterns identified in RQ1 be operationalised into an automated system that, given an arbitrary new date range, identifies the most similar historical regime and deploys an optimally weighted model ensemble?

1.3 Project Evolution

This project extends the interim study [11] in which baseline point-to-point prediction was established using ARIMA, LSTM, and GRU on NASDAQ and NYSE stocks. Key findings from the interim study included that GRU consistently outperforms LSTM (achieving 25–42% lower MAPE) and that ARIMA remains competitive for stable, trending stocks. The final project substantially expands upon this foundation:

- **Model scope:** Expanded from 3 to 7 architectures, adding bidirectional variants, PatchTST, Mamba, and SNN.
- **Market scope:** Shifted from individual stocks to index-level prediction on SP500 and HSI.
- **Evaluation framework:** Replaced simple train-test splits with a walk-forward protocol across 7 special periods, 5 window sizes, and 3 forecast horizons.
- **Novel contribution:** Designed and implemented the Smart Prediction module for automated regime detection and model selection.
- **Analysis tooling:** Developed `Analyzer.py` for post-hoc analysis across multiple experimental seeds.

1.4 Research Objectives

1. Implement and evaluate seven time-series forecasting architectures on SP500 and HSI.
2. Conduct systematic special-period analysis across seven distinct market regimes.
3. Quantify the effects of forecast horizon and lookback window on prediction accuracy.
4. Compare predictability between SP500 and HSI, identifying structural explanatory factors.
5. Design, implement, and validate a Smart Prediction module for automated regime matching and model selection.
6. Provide practical recommendations for model deployment under different market conditions.

2. Literature Review

2.1 Classical Statistical Methods

The ARIMA (AutoRegressive Integrated Moving Average) model, formalised by Box and Jenkins [1], remains the dominant classical baseline for univariate time series forecasting. By combin-

ing autoregressive terms, differencing for stationarity, and moving average components, ARIMA directly models the autocorrelation structure of price series. For short-horizon financial prediction, this autocorrelation structure is the strongest available signal [2], which explains ARIMA's persistent competitiveness. The M4 Competition [14] notably demonstrated that simple statistical methods frequently match or outperform complex neural networks on many forecasting tasks.

2.2 Recurrent Deep Learning Architectures

Long Short-Term Memory (LSTM) networks [9] addressed the vanishing gradient problem through a three-gate mechanism, enabling models to capture long-range temporal dependencies. Empirical studies in financial forecasting have shown LSTM can outperform traditional methods during volatile periods [7]. However, the additional complexity of LSTM's three gates compared to the simpler two-gate GRU [3] often leads to overfitting in limited financial datasets. This project uses bidirectional variants of all three recurrent architectures, enabling the model to process sequences in both temporal directions.

2.3 Transformer-Based Models: PatchTST

PatchTST [16] applies the Transformer architecture to time series by dividing input sequences into non-overlapping patches and treating each patch as a token for self-attention. Designed for long-range multivariate forecasting, PatchTST has achieved strong performance on standard benchmarks. However, its patch-based tokenisation inherently coarsens temporal resolution, which — as demonstrated in this study — proves detrimental for financial prediction tasks driven primarily by short-range autocorrelations.

2.4 State-Space Models: Mamba

The Mamba model [8] introduces a selective state-space mechanism where the transition parameters (Δ, B, C) are input-dependent, enabling the model to dynamically decide which historical context to retain. This selectivity is particularly well-matched to financial markets that exhibit both long-range macro-trend dependencies and short-range momentum effects. The continuous-time formulation also provides a principled alternative to attention-based models for sequential data.

2.5 Spiking Neural Networks

Spiking Neural Networks (SNNs) [13] use binary spike-based computation inspired by biological neural processing. While energy-efficient in hardware implementations, their binary discretisation of continuous financial signals introduces quantisation noise that, as found in this study, substantially degrades prediction quality across most market regimes.

2.6 Financial Time Series and Market Regimes

Financial returns exhibit well-documented properties relevant to model selection: volatility clustering [15], non-stationarity with structural breaks [2], and varying degrees of efficiency across markets. Research on the HSI has stronger autocorrelation and lower efficiency compared to the SP500, attributable to higher retail investor participation and periodic regulatory interventions. These structural differences might be the cross-market comparison central to this study.

3. Methodology

3.1 Data Sources

The data across the US stock market and the Hong Kong stock market.

S&P 500 (SP500): Daily OHLCV data for the ˆGSPC ticker covering 2000-01-01 to 2025-01-01 (≈6,300 trading days), sourced via the `yfinance` library with local CSV caching (`GSPC.csv`).

Hang Seng Index (HSI): Daily OHLCV data for the ˆHSI ticker over the same date range (≈5,900 trading days), sourced analogously (`HSI.csv`).

3.2 Feature Engineering

A 7-dimensional feature vector is constructed daily from raw OHLCV data, providing the model with information about momentum, volatility, and microstructure:

Table 3.1: Feature Engineering Pipeline

Feature	Formula	Rationale
Log Return	$\ln(P_t/P_{t-1})$	Stationary return proxy; primary predictive signal
HL Percentage	$(H_t - L_t)/C_t$	Intra-day volatility
OC Percentage	$(C_t - O_t)/O_t$	Opening-to-close direction
RSI (14-day)	Standard formula	Momentum oscillator
MACD	12/26/9 EMA difference	Trend-following momentum
ATR (14-day)	Average True Range	Volatility and risk measure
Normalised Close	Min-Max $\in [0, 1]$	Prediction target (last column)

3.3 Walk-Forward Training Protocol

A strict walk-forward (expanding-window) methodology is applied throughout. For each special period (t_s, t_e):

1. **Training data:** All observations before t_s.
2. **Validation data:** Last 10% of training data (for early stopping with patience = 15 epochs).
3. **Test data:** Period $[t_s, t_e]$ with a lookback buffer of `seq_len` days before t_s.

This ensures zero information leakage from the evaluation period into model training, making results comparable to genuine out-of-sample prediction.

3.4 Evaluation Metrics

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{3.1}$$

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{3.2}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \tag{3.3}$$

The coefficient of determination (R^2) is the primary metric for cross-condition comparison due

to its scale-invariance: $R^2 = 0$ indicates performance equivalent to predicting the mean; negative R^2 indicates worse-than-mean predictions.

3.5 Experimental Design

Table 3.2: Predefined Special Periods

Period	Date Range	Description
2008 Financial Crisis	2007-10 to 2009-03	Subprime collapse, systemic bank failures
2010 Euro Debt Crisis	2010-04 to 2012-06	Sovereign debt crisis, ECB interventions
2015 China Crash	2015-06 to 2016-02	China equity bubble burst and global spillover
2018 Trade War	2018-01 to 2018-12	US-China tariff escalation
2020 COVID Crash	2020-02 to 2020-06	Pandemic crash and V-shaped recovery
2022 Rate Hike	2022-01 to 2022-12	Fed aggressive tightening cycle
2023 AI Wave	2023-01 to 2023-12	AI-driven bull market (ChatGPT, NVIDIA)

The full experimental grid covers 2 markets $\times$ 8 models $\times$ 7 periods $\times$ 5 windows $\times$ 3 horizons = 1,680 model evaluations.

4. System Design and Implementation

4.1 Overall Architecture

The project is implemented in two primary Python modules:

- **`SpecialPeriod.py`** ($\approx$62k bytes): Main prediction engine with Classic Mode and Smart Predict Mode.
- **`Analyzer.py`** ($\approx$27k bytes): Post-hoc analysis and visualisation across multiple experimental seeds.

4.2 Model Implementations

4.2.1 ARIMA

Implemented via `statsmodels` with order $(2, 1, 2)$. For each target date in the evaluation period, ARIMA is re-fitted on the preceding `seq_len` normalised close prices and used to forecast `horizon` steps ahead — reflecting a realistic rolling-window deployment scenario.

4.2.2 Bidirectional Recurrent Models

All three bidirectional models (Bi-RNN, Bi-GRU, Bi-LSTM) share the `BaseRecurrent` architecture: a bidirectional recurrent layer (256 units per direction) followed by a linear output head. Four recurrent layers with 0.2 dropout are used throughout.

4.2.3 PatchTST

Input sequences are divided into patches of length 8 with stride 4, projected into a $d_{model} = 256$ embedding space with learnable positional encodings, then processed by a 4-layer Transformer encoder with 4 attention heads. The concatenated patch outputs are mapped to a scalar prediction.

4.2.4 Mamba State-Space Model

A custom implementation of the selective SSM architecture:

$$h_t = \bar{A}h_{t-1} + \bar{B}x_t, \quad y_t = C_t h_t + Dx_t \tag{4.1}$$

where $\bar{A} = \exp(\Delta A)$, $\bar{B} = \Delta B$, and Δ, B, C are input-dependent projections (the selective mechanism). Configuration: $d_{\text{state}} = 32$, $d_{\text{conv}} = 4$, expansion factor $= 2$.

4.2.5 Spiking Neural Network

A two-layer feedforward network with Leaky ReLU activations approximating integrate-and-fire behaviour. Temporal aggregation is performed by mean-pooling across the time dimension.

4.3 Training Configuration

Table 4.1: Shared Training Hyperparameters

Hyperparameter	**Value**
Optimiser	AdamW
Learning rate	3×10^{-4}
Loss function	Smooth L1 (Huber loss)
Batch size	64
Max epochs	100
Early stopping patience	15 epochs
Hidden dimension	256
Number of layers	4
Dropout	0.2
Validation split	10% of training data

4.4 Analyzer Module

`Analyzer.py`: multiseed aggregation for variance estimation; period difficulty and model rankings; horizon degradation curves; window sensitivity analysis; cross-market comparison; and

publication-ready heatmap and chart generation.

5. Data Collection

5.1 Dataset Characteristics

Table 5.1: Special Period Sample Sizes

Period	SP500 Days	HSI Days	Difficulty
2008 Financial Crisis	378	367	Easiest
2010 Euro Debt Crisis	568	555	Easy
2015 China Crash	189	186	Medium
2018 Trade War	251	246	Medium
2020 COVID Crash	104	102	Hardest
2022 Rate Hike	251	246	Medium-Hard
2023 AI Wave	250	243	Medium

The 2020 COVID period presents the most challenging scenario with only 104 trading days on SP500, severely constraining model learning in addition to the extreme non-stationarity of the price dynamics.

5.2 Data Quality and Preprocessing

Missing values are forward-filled for gaps of up to 5 consecutive trading days. RSI and ATR indicators require a 14-day lookback window, so the first 14 rows after any data gap are dropped. Min-Max scaling to $[0, 1]$ is applied jointly across the 7 features, fitted on training data only to prevent leakage. No outlier removal is applied: extreme returns during crisis periods are precisely

the dynamics under study, and removing them would misrepresent the evaluation conditions.

6. SP500 Experimental Results

6.1 Period Difficulty Ranking

Figure 6.1 and Table 6.1 present the average R^2 across all models, windows, and horizons for each special period.

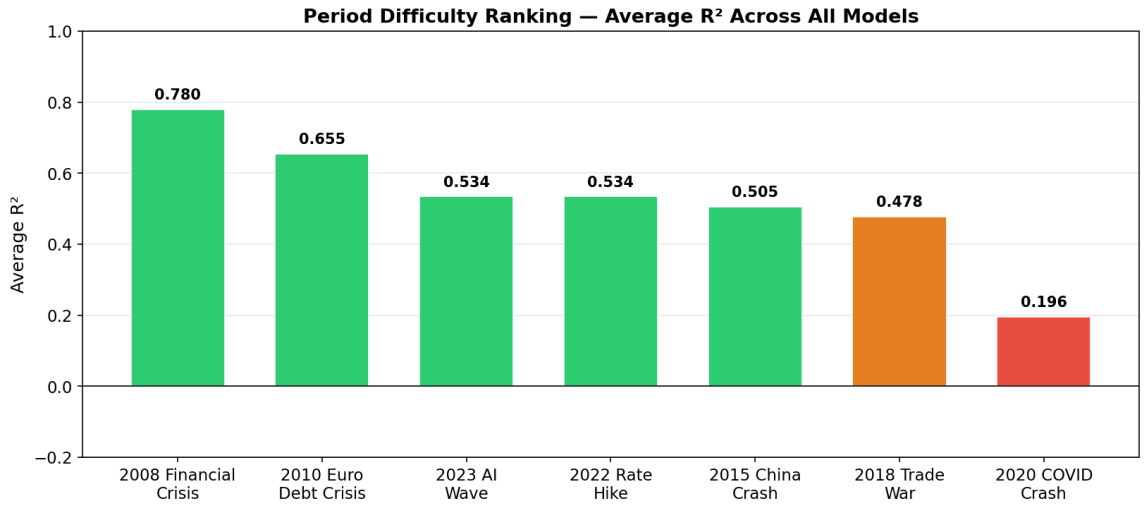

Figure 6.1: SP500 Period Difficulty Ranking — Average R^2 across all models, window sizes, and forecast horizons. Green bars indicate high predictability (trend-driven periods); red bars indicate low predictability (event-driven or black-swan periods).

Table 6.1: SP500 Period Difficulty Ranking (Average R^2 across All Models, Windows, and Horizons)

Rank	Period	Avg R^2	Avg RMSE	Avg MAE	Samples
1	2008 Financial Crisis	0.7268	0.0191	0.0144	378
2	2010 Euro Debt Crisis	0.5395	0.0096	0.0077	568
3	2023 AI Wave	0.3604	0.0277	0.0235	250
4	2022 Rate Hike	0.2723	0.0381	0.0321	251
5	2015 China Crash	0.2145	0.0117	0.0091	189
6	2018 Trade War	0.1983	0.0155	0.0119	251
7	2020 COVID Crash	0.0683	0.0438	0.0318	104

The most striking finding is that the 2008 Financial Crisis — the most severe systemic event — is paradoxically the easiest to predict (R^2=0.73). The crisis followed a sustained, strongly autocorrelated downtrend driven by identifiable macro events (subprime collapse, Lehman Brothers

failure, TARP announcements), which is precisely the structure that time-series models exploit. In contrast, the 2020 COVID Crash (R^2=0.07) is hardest: the 34% decline in 23 trading days followed by an equally rapid V-shaped recovery created an unprecedented, non-stationary pattern with no historical analogue.

6.2 Model Ranking

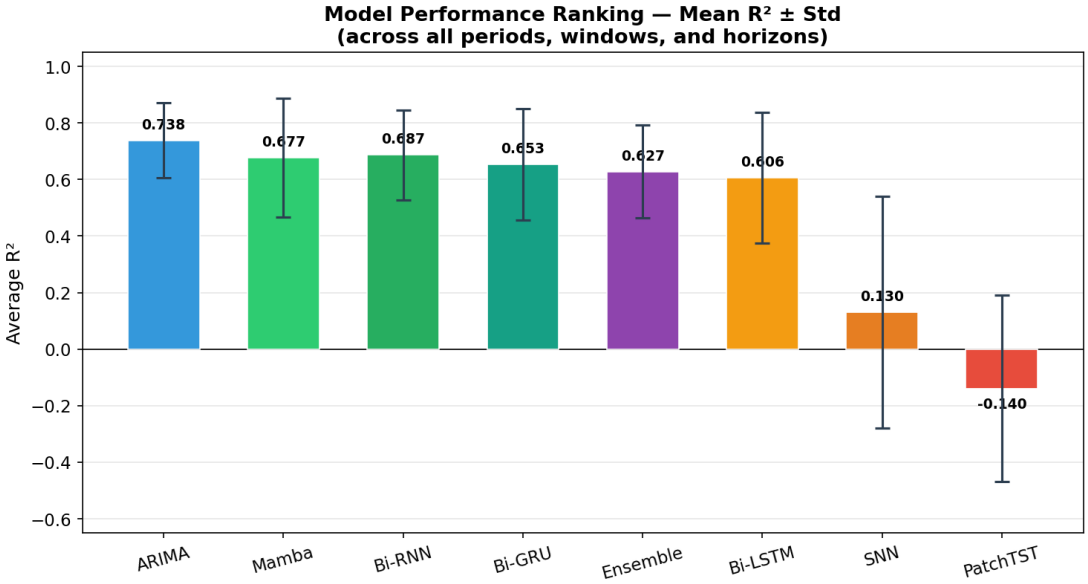

Figure 6.2: SP500 Model Performance Ranking — Mean $R^2 \pm$ standard deviation across all periods, windows, and horizons. Error bars represent inter-condition variability (lower = more stable).

Table 6.2: SP500 Model Overall Ranking

Rank	Model	Avg R^2	Avg RMSE	Avg MAE	Std(R^2)
1	ARIMA	0.7501	0.0146	0.0112	0.1645
2	Mamba	0.6367	0.0177	0.0137	0.2937
3	Bi-RNN	0.6303	0.0187	0.0145	0.1724
4	Bi-GRU	0.5827	0.0189	0.0141	0.2517
5	Ensemble	0.5687	0.0211	0.0161	0.1906
6	Bi-LSTM	0.4798	0.0211	0.0163	0.3632
7	SNN	−0.1282	0.0342	0.0279	0.6969
8	PatchTST	−0.7999	0.0427	0.0355	0.7439

Key observations:

ARIMA dominance. ARIMA achieves the highest R^2 in every single period. Its autoregressive structure directly exploits serial autocorrelation in daily prices. Its low Std(R^2) = 0.16 confirms robustness across diverse market regimes.

Mamba as best deep learning model. Mamba's selective state-space mechanism dynamically adjusts which historical context to retain, matching financial markets that exhibit both trend-following and mean-reversion behaviour.

PatchTST catastrophic failure. PatchTST achieves negative R^2 in 6 of 7 periods, this might confirming that patch-based tokenisation is fundamentally unsuitable for short-range autocorrelation-dominated financial prediction.

Ensemble underperformance. The equal-weight ensemble (R^2=0.57) underperforms both Mamba (0.64) and Bi-RNN (0.63), demonstrating that naive ensembling is harmful when poorly performing constituents (PatchTST, SNN) contaminate the aggregate.

Figure 6.3 provides a qualitative illustration of these rankings via actual vs. predicted price curves for all seven models on the SP500 2008 Financial Crisis period at W=252, H=10 — the most demanding horizon, where architectural differences are most visually pronounced.

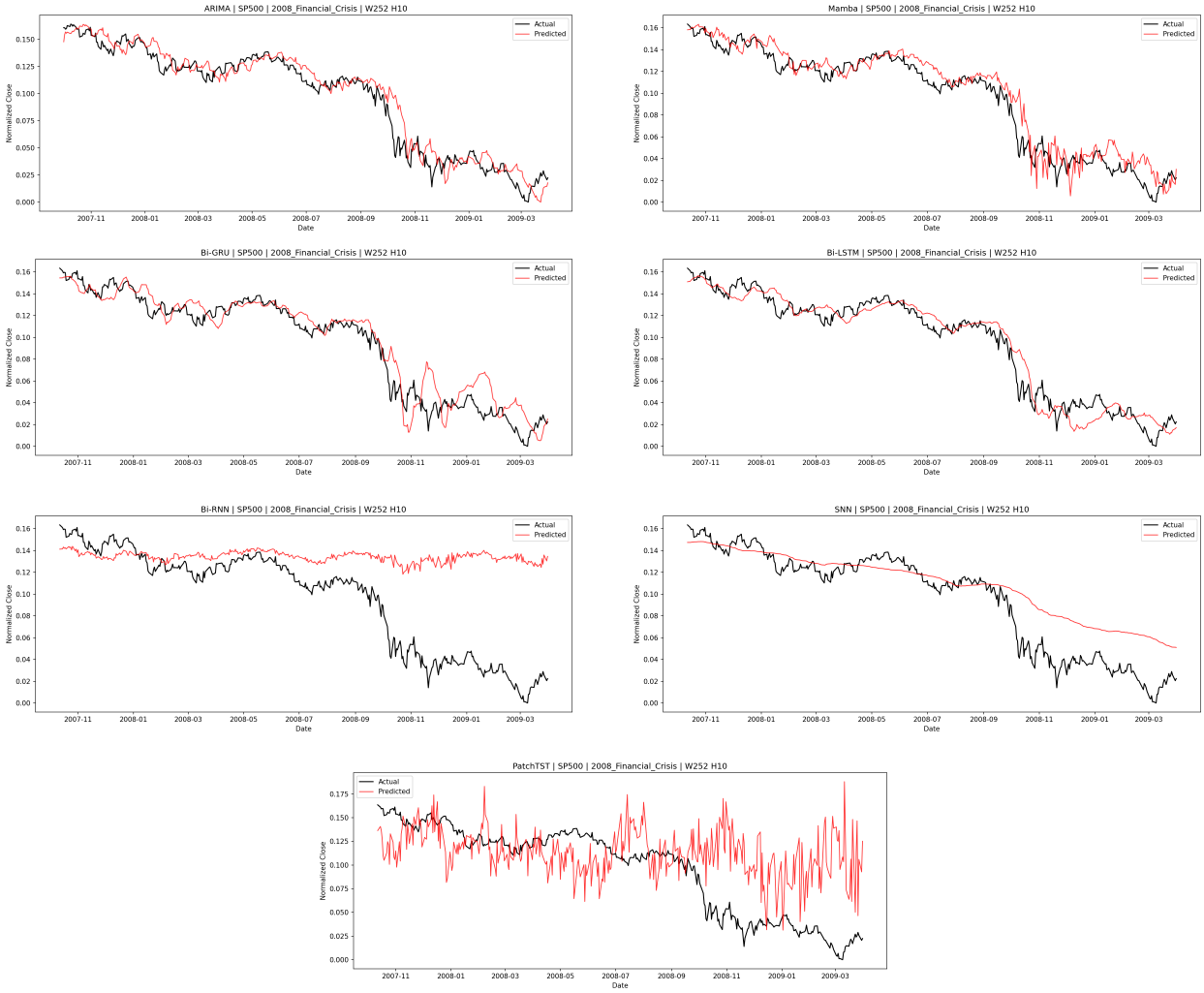

Figure 6.3: Actual vs. predicted normalised closing price for all seven models on SP500, 2008 Financial Crisis, W=252, H=10. Top row: ARIMA (left) and Mamba (right) both track the sustained downtrend closely, with ARIMA maintaining the tightest alignment through the acute October 2008 shock. Middle row: Bi-GRU and Bi-LSTM follow the general trend but introduce a systematic positive bias in the post-crash stabilisation phase (Nov 2008 – Mar 2009). Bottom-left: Bi-RNN collapses to a near-flat prediction around 0.13–0.14, reverting to a mean-like output at this horizon despite its competitive aggregate R^2. Bottom-middle: SNN tracks the direction in a compressed, lagged fashion, overestimating prices after the October shock. Bottom-right: PatchTST produces erratic high-frequency oscillations with no directional coherence, visually confirming its architectural incompatibility with short-range financial forecasting.

6.3 Analysis by Prediction Horizon

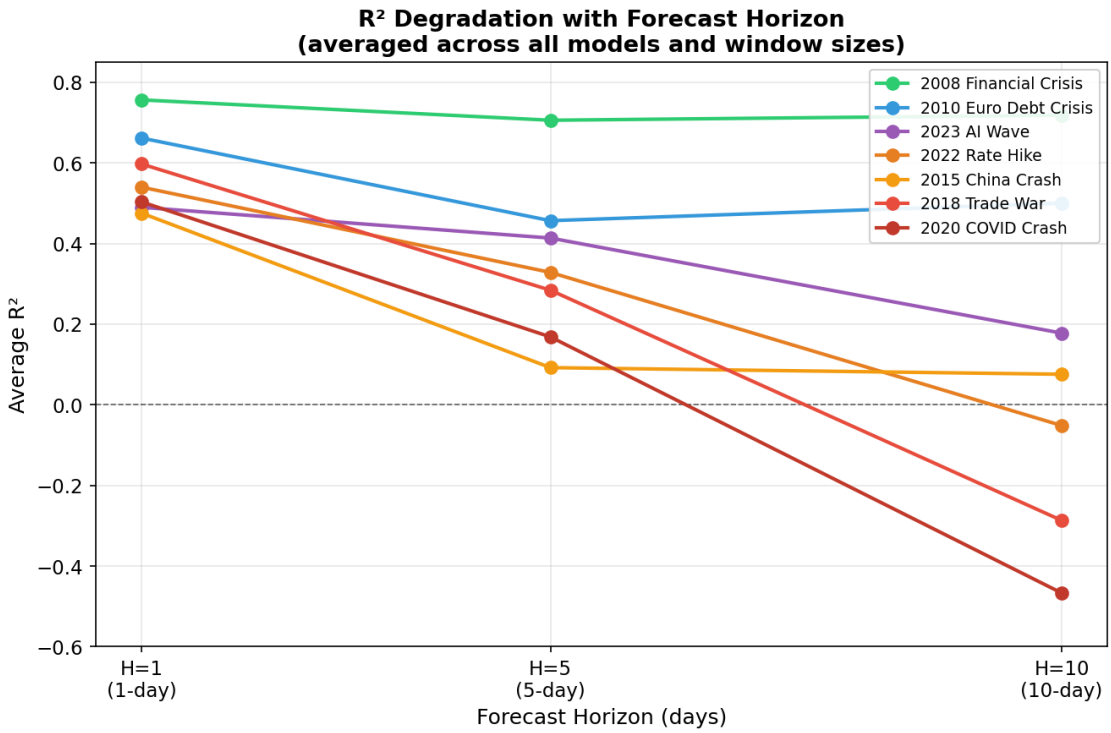

Figure 6.4: R² degradation with forecast horizon for each special period. Each line shows the average R² across all models and window sizes. The 2008 Financial Crisis is uniquely horizon-resilient (dashed green), while COVID, Trade War, and Rate Hike collapse to negative R² at H=10.

Table 6.3: SP500 R² by Period and Forecast Horizon

Period	H=1	H=5	H=10	Degradation
2008 Financial Crisis	0.7563	0.7058	0.7182	−5.0%
2010 Euro Debt Crisis	0.6617	0.4567	0.5002	−24.4%
2023 AI Wave	0.4899	0.4136	0.1778	−63.7%
2022 Rate Hike	0.5399	0.3284	−0.0514	−109.5%
2015 China Crash	0.4756	0.0922	0.0758	−84.1%
2018 Trade War	0.5974	0.2842	−0.2868	−148.0%
2020 COVID Crash	0.5034	0.1684	−0.4668	−192.7%
Overall average	**0.5749**	**0.3499**	**0.0953**	**−83.4%**

Forecast horizon is the *single most influential factor*: R² declines 83% from H=1 to H=10, dwarfing the effects of model selection or window size. The 2008 Financial Crisis exhibits remarkable horizon resilience (−5%), because the sustained directional momentum of the crash persisted across

multi-day windows. In contrast, the COVID crash, Trade War, and Rate Hike all collapse to negative R^2 at H=10.

6.4 Analysis by Window Size

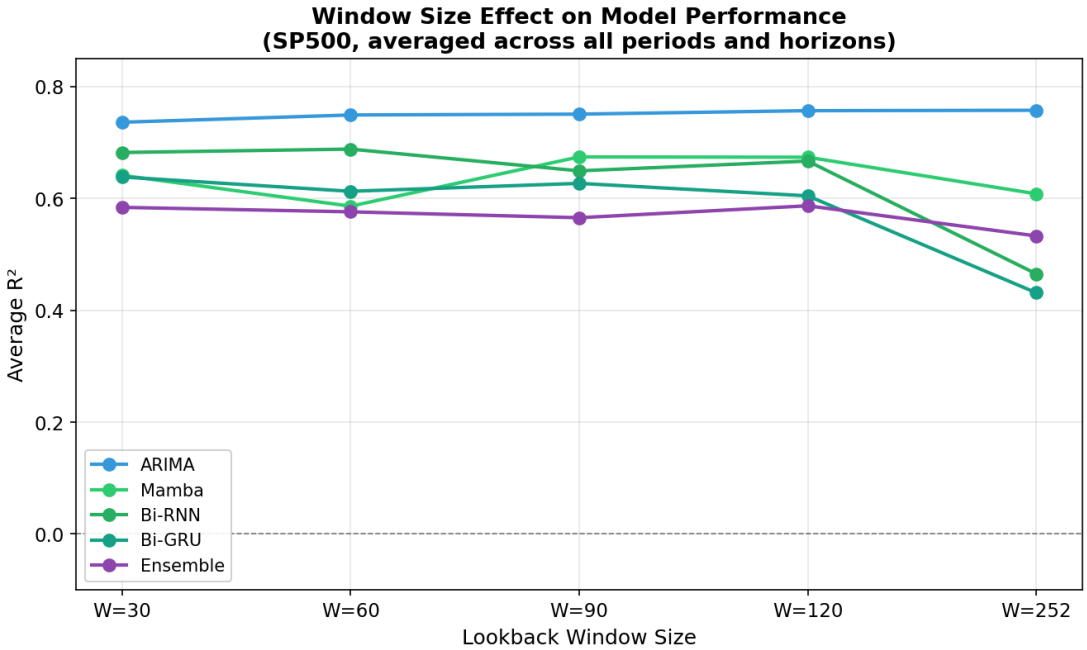

Figure 6.5: Effect of lookback window size on R^2 for key models (SP500, averaged across all periods and horizons). ARIMA uniquely benefits from longer windows; deep learning models generally prefer W=30–120.

Table 6.4: SP500 Performance by Lookback Window

Window (Days)	Avg R^2	Avg RMSE
30	0.3769	0.0224
60	0.3585	0.0237
90	0.3218	0.0239
120	0.3829	0.0229
252	0.2600	0.0253

Window size has a minor effect relative to horizon and model choice. W=120 (6 months, covering two earnings cycles) is marginally optimal for deep learning models. ARIMA uniquely improves monotonically with window size (W=30: 0.736, W=252: 0.758), as more data improves AR coefficient estimation. Deep learning models, particularly Bi-RNN, degrade severely at W=252 (0.68 $\rightarrow$ 0.47), as a full year of lookback introduces stale data from different market regimes.

6.5 R² Heatmap — Models × Periods

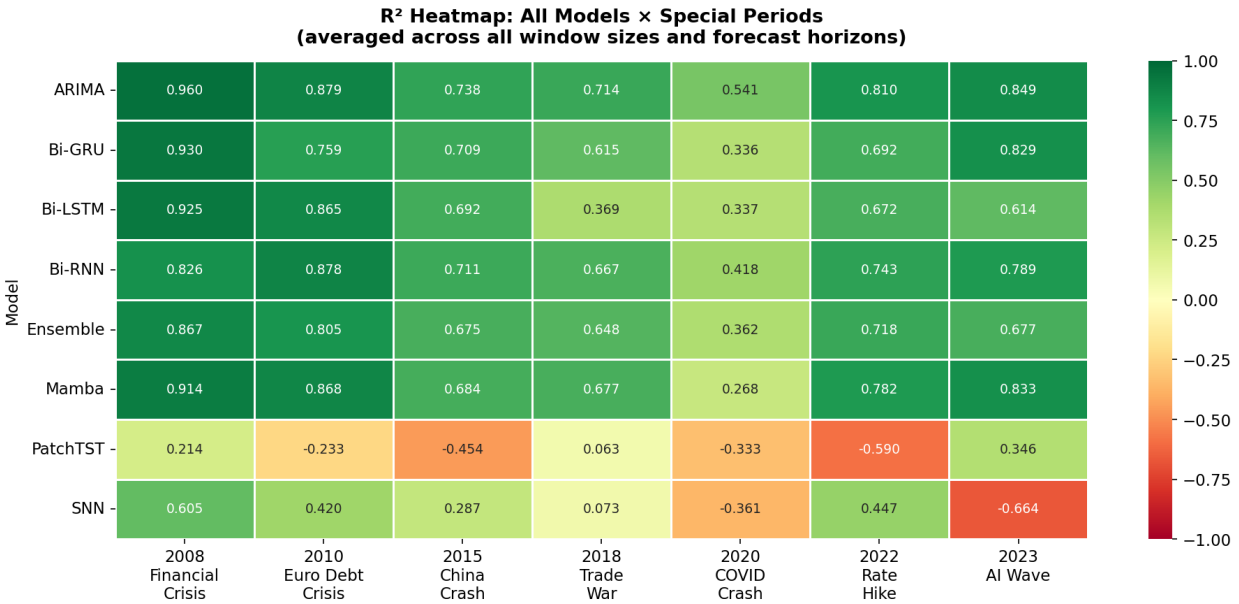

Figure 6.6: SP500 R² Heatmap: All models × all special periods, averaged across all window sizes and forecast horizons. Green = high predictability, red/white = poor performance, values < 0 indicate worse than mean prediction.

Table 6.5: SP500 Average R² by Model and Special Period

Model	2008	2010	2015	2018	2020	2022	2023
ARIMA	0.9602	0.8788	0.7383	0.7139	0.5411	0.8098	0.8491
Mamba	0.9140	0.8683	0.6844	0.6772	0.2682	0.7816	0.8334
Bi-RNN	0.8256	0.8781	0.7106	0.6671	0.4178	0.7433	0.7886
Bi-GRU	0.9305	0.7591	0.7093	0.6147	0.3361	0.6917	0.8294
Ensemble	0.8665	0.8054	0.6746	0.6480	0.3618	0.7180	0.6773
Bi-LSTM	0.9248	0.8648	0.6922	0.3690	0.3374	0.6720	0.6135
SNN	0.6054	0.4196	0.2869	0.0730	−0.3611	0.4466	−0.6645
PatchTST	0.2142	−0.2326	−0.4538	0.0629	−0.3334	−0.5895	0.3463

6.6 COVID Crash — Detailed Horizon Analysis

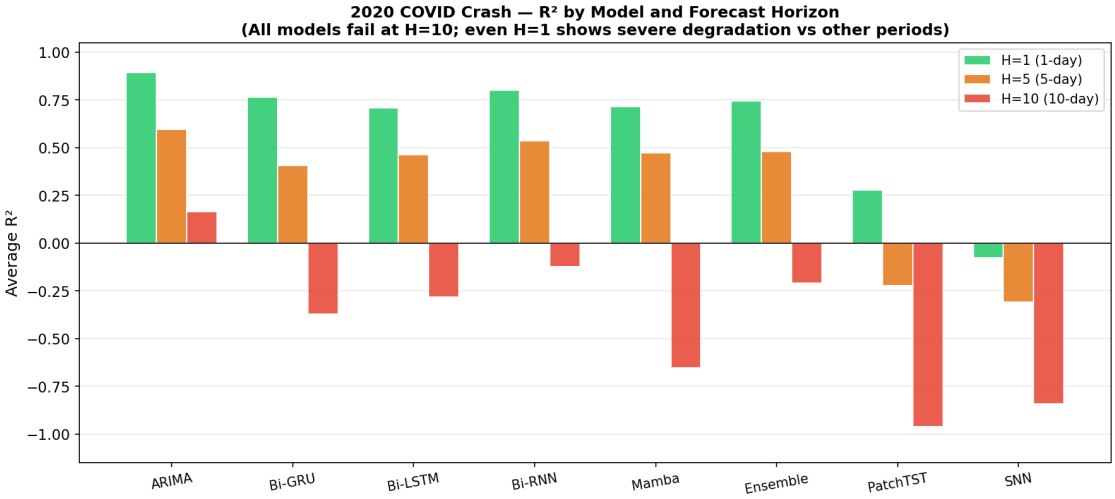

Figure 6.7: 2020 COVID Crash — R^2 by model and forecast horizon. At H=10, all models achieve negative R^2, performing worse than a constant mean prediction. This is the clearest demonstration that black swan events are fundamentally unpredictable with historical price data alone.

The COVID crash result (Figure 6.7) is the most important finding of the entire study: *no model, regardless of architecture, can reliably predict 10-day-ahead prices during an unprecedented black swan event*. The R^2 values ranging from -0.37 (Ensemble) to -0.96 (PatchTST) confirm this. Even at H=1, the best models only achieve $R^2 \approx 0.91$ — significantly below their performance in other periods — because the V-shaped recovery created daily reversals with no historical precedent.

6.7 Cross-Cutting Insights

6.7.1 Trend-Driven vs. Event-Driven Periods

Table 6.6: SP500 Performance by Market Regime Type

Type	Periods	Avg R^2	Characteristics
Trend-driven	2008, 2010, 2023	0.54	Sustained directional momentum
Event-driven	2018, 2020, 2015	0.16	Exogenous shocks, regime shifts
Mixed	2022	0.27	Policy-driven with structural breaks

Models perform approximately 3.4× better (by R^2) in trend-driven vs. event-driven periods. This is consistent with the EMH: trend-driven periods exhibit stronger serial correlation that statistical models can exploit, while event-driven periods approximate random walk behaviour.

7. HSI Results and Cross-Market Comparison

7.1 HSI Overall Performance

Table 7.1: HSI Period Difficulty Ranking

Rank	Period	Avg R^2	Avg RMSE	Avg MAE	Samples
1	2008 Financial Crisis	0.8335	0.0812	0.0661	367
2	2022 Rate Hike	0.7961	0.0392	0.0309	246
2	2015 China Crash	0.7961	0.0377	0.0295	186
4	2010 Euro Debt Crisis	0.7709	0.0319	0.0253	555
5	2018 Trade War	0.7582	0.0394	0.0323	246
6	2023 AI Wave	0.7079	0.0280	0.0224	243
7	2020 COVID Crash	0.3237	0.0433	0.0322	102

The most striking HSI result is the 2015 China Crash: while SP500 found this period among the most difficult (R^2=0.21), HSI achieves R^2=0.80 — a $+0.58$ improvement. The explanation is straightforward: HSI was a direct participant in the Chinese market turmoil through the Stock Connect programme, creating structured, learnable price dynamics. The SP500 only experienced indirect contagion through global capital flows, generating noisy and unpredictable spillover patterns.

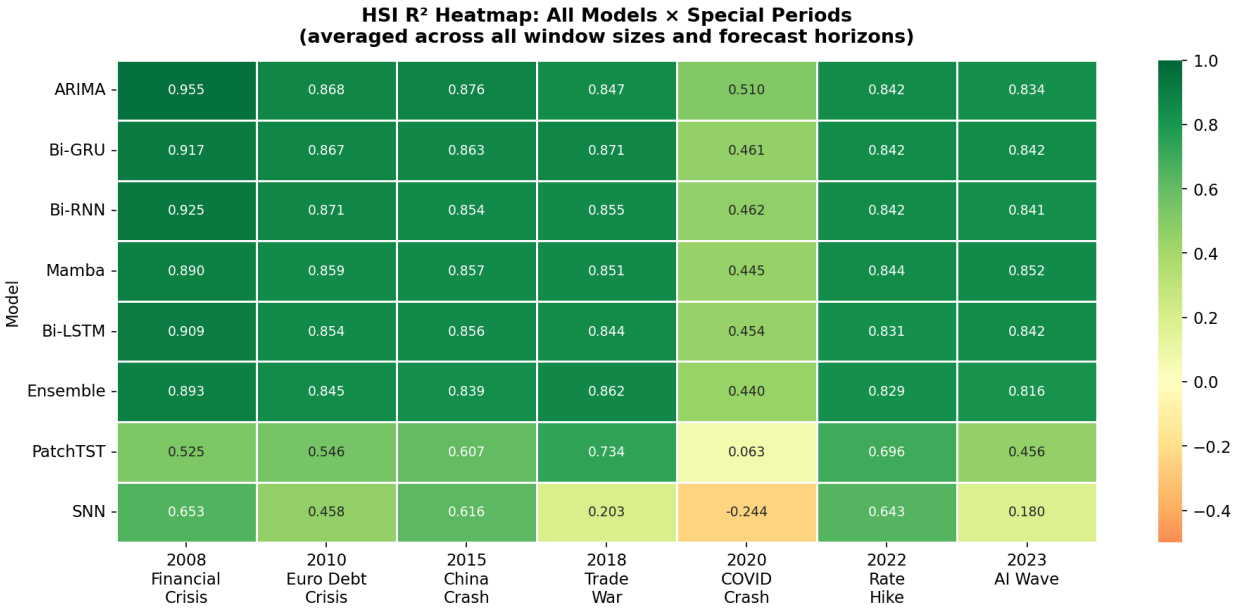

Figure 7.1: HSI R² Heatmap: All models × all special periods. Note the dramatically tighter range compared to SP500 (Figure 6.6) — even PatchTST and SNN achieve positive R² across most periods on HSI.

Table 7.2: HSI Model Overall Ranking

Rank	Model	Avg R²	Avg RMSE	Avg MAE	Std(R²)
1	ARIMA	0.8188	0.0329	0.0256	0.1421
2	Bi-GRU	0.8090	0.0343	0.0272	0.1556
3	Bi-RNN	0.8071	0.0342	0.0269	0.1551
4	Mamba	0.7996	0.0361	0.0283	0.1572
5	Bi-LSTM	0.7984	0.0360	0.0283	0.1540
6	Ensemble	0.7890	0.0379	0.0295	0.1559
7	PatchTST	0.5182	0.0627	0.0511	0.2229
8	SNN	0.3585	0.0694	0.0561	0.3325

The HSI top-6 models cluster in a remarkably narrow range of 0.79–0.82 R², compared to the 0.48–0.75 spread on SP500. Even PatchTST achieves positive R²=0.52 on HSI (vs. −0.80 on SP500), and SNN achieves 0.36 (vs. −0.13), confirming that HSI's stronger signal-to-noise ratio is a market characteristic that benefits all architectures.

7.2 Cross-Market Comparison

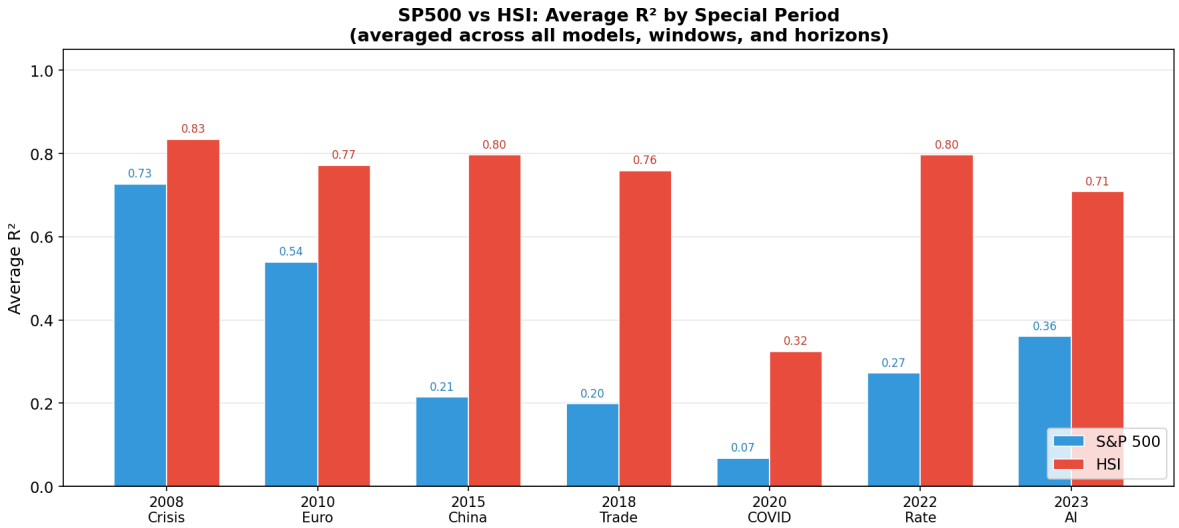

Figure 7.2: SP500 vs. HSI average R² by special period. The HSI consistently and substantially outperforms SP500 across all periods. The largest gap is the 2015 China Crash (+0.58), where HSI was a direct participant while SP500 experienced only indirect spillover.

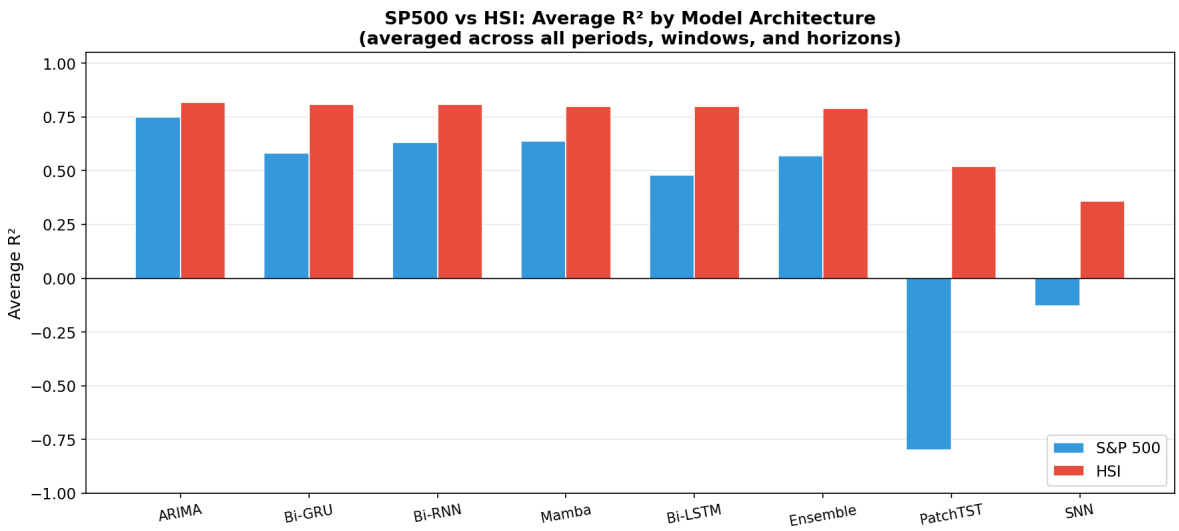

Figure 7.3: SP500 vs. HSI average R² by model architecture. HSI outperforms SP500 for every model. The largest improvement is for PatchTST (+1.32), confirming that its architectural requirements (long-range autocorrelation) are better met by HSI.

Table 7.3: SP500 vs. HSI: R² Improvement by Period and Model

	By Period		By Model	
	Period	**Improvement**	**Model**	**Improvement**
Largest	2015 China Crash	+0.58	PatchTST	+1.32
	2018 Trade War	+0.56	SNN	+0.49
	2022 Rate Hike	+0.52	Bi-LSTM	+0.32
	2023 AI Wave	+0.35	Bi-GRU	+0.23
	2020 COVID	+0.26	Ensemble	+0.22
	2010 Euro Debt	+0.23	Bi-RNN	+0.18
Smallest	2008 Financial	+0.11	ARIMA	+0.07

The visual contrast between markets is illustrated in Figure 7.4, which shows the same seven models on HSI under identical conditions (W=252, H=10, 2008 Financial Crisis). Compared to Figure 6.3, all models exhibit tighter predicted-actual alignment, reflecting HSI's stronger autocorrelation signal. Most notably, Bi-RNN — which collapsed to a flat line on SP500 — produces a coherent downtrend on HSI, as the larger normalised price range (0.10–0.93 vs. 0.00–0.16) provides a stronger gradient signal at H=10. PatchTST also partially recovers directional coherence on HSI, though with a persistent positive offset, consistent with its $+1.32$ R² gain in Table 7.3. The Mamba curve shows a transient spike to -0.2 at peak volatility in November 2008 — a numerical instability absent on SP500 — suggesting that gradient clipping may be warranted for HSI under extreme regimes.

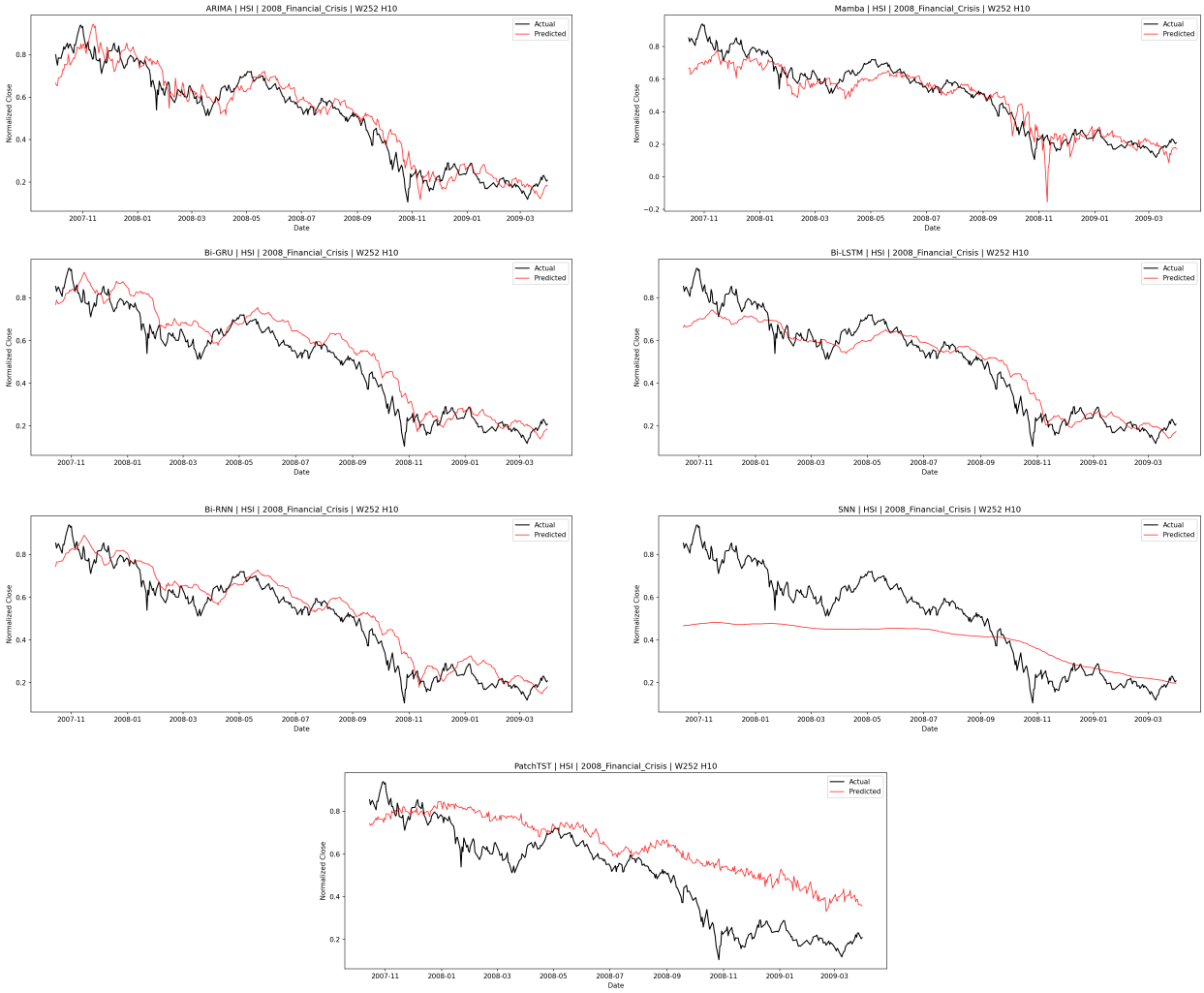

Figure 7.4: Actual vs. predicted normalised closing price for all seven models on HSI, 2008 Financial Crisis, W=252, H=10 (cf. Figure 6.3 for SP500). All models demonstrate visibly tighter tracking than their SP500 counterparts. ARIMA and the bidirectional recurrent models follow the full 0.93–0.10 price range closely; Bi-RNN in particular recovers coherent trend-following behaviour unlike its flat SP500 prediction. Mamba exhibits a transient negative spike in November 2008. PatchTST achieves a partially directional trajectory with a positive offset, in contrast to its erratic SP500 oscillations. SNN remains amplitude-suppressed throughout.

7.3 Why HSI is More Predictable

Five structural factors explain HSI's superior predictability:

1. **Stronger autocorrelation** from Chinese market influences, retail investor dominance, and capital controls.
2. **Direct event exposure** — HSI was a primary participant in the China Crash and Trade War, creating structured price reactions.
3. **Market microstructure** — overlap with multiple Asian markets creates predictable information catch-up patterns.

4. **USD peg dynamics** — Hong Kong's currency peg mechanically transmits US monetary policy to HK asset prices.
5. **Lower efficiency** — higher retail participation and periodic regulatory interventions create exploitable inefficiencies.

Deep learning models benefit more from the HSI's stronger signal (Bi-LSTM: +0.32 R^2) than ARIMA does (+0.07), suggesting that HSI's price dynamics contain nonlinear patterns that deep learning can exploit but that ARIMA's linear structure cannot.

8. Smart Prediction Module

8.1 Motivation

Classic Mode requires domain expertise to select appropriate models for a given market regime. The Smart Prediction module automates the full workflow: given only a start date, end date, and market, the system identifies the regime, selects the best weighted model ensemble, and generates predictions — requiring zero manual model selection.

8.2 System Architecture

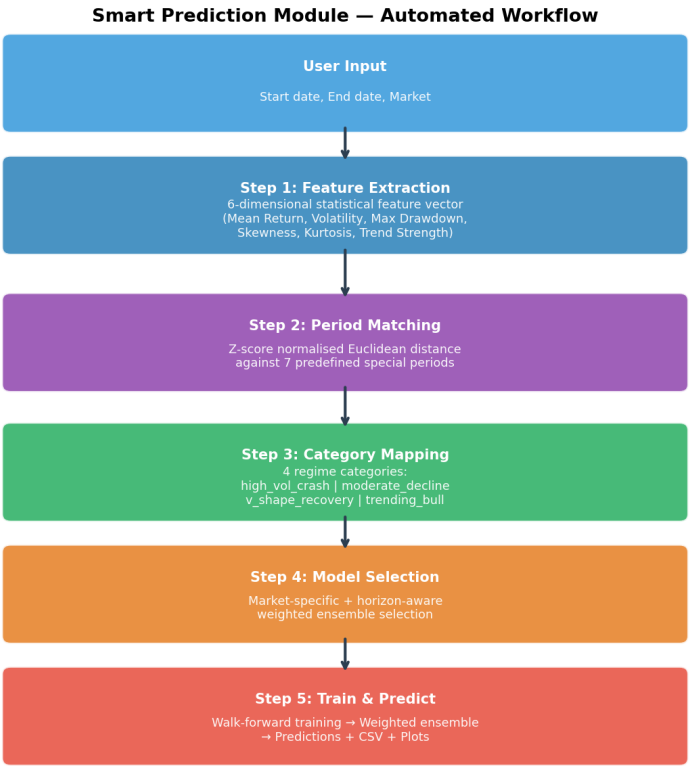

Figure 8.1: Smart Prediction module pipeline. The five-step workflow proceeds from user input to prediction output, with regime matching driving automated model selection.

8.3 Feature Extraction

A 6-dimensional statistical feature vector is extracted from the actual market data in the input date range:

Table 8.1: Smart Prediction Feature Vector

#	Feature	Formula	Interpretation
1	Mean Return	$\overline{r_t}$	Distinguish bull/bear/sideways
2	Volatility	$\sigma(r_t)$	Separate crises from calm trends
3	Max Drawdown	$\max(\text{peak} - P)/\text{peak}$	Severity of price decline
4	Skewness	$\text{skew}(r_t)$	Left-skew = frequent sharp drops
5	Kurtosis	$\text{kurt}(r_t)$	Fat tails / black swan events
6	Trend Strength	$\sum r_t/\sigma(r_t)$	Sharpe-like directional conviction

8.4 Period Matching Algorithm

Algorithm 1 Z-Score Normalised Euclidean Period Matching

Require: Input feature vector $\mathbf{f}_{\text{input}} \in \mathbb{R}^6$; reference vectors $\{\mathbf{f}_k\}_{k=1}^{7}$

Stack: $\mathbf{F} = [\mathbf{f}_1; \ldots; \mathbf{f}_7; \mathbf{f}_{\text{input}}] \in \mathbb{R}^{8\times 6}$

Joint z-score normalise: $\tilde{\mathbf{F}}_{ij} = (F_{ij} - \mu_j)/\sigma_j$ (per feature j)

for $k = 1$ to 7 **do**

    $d_k = \|\tilde{\mathbf{f}}_{\text{input}} - \tilde{\mathbf{f}}_k\|_2$

end for

return Periods ranked by ascending d_k; best match $= \arg\min_k d_k$

Joint z-score normalisation ensures all six features contribute equally regardless of their original scale, preventing high-variance features (e.g. kurtosis) from dominating the distance metric.

8.5 Category Mapping

Table 8.2: Regime Category Mapping

Category	Mapped Periods	Key Characteristics
`high_vol_crash`	2008 Crisis, 2020 COVID	Extreme volatility, deep drawdown
`moderate_decline`	2010 Euro, 2018 Trade War, 2022 Rate	Gradual decline, policy-sensitive
`v_shape_recovery`	2015 China Crash	Sharp drop + rapid rebound
`trending_bull`	2023 AI Wave	Sustained uptrend, moderate vol.

8.6 Model Selection Strategy

8.6.1 Two-Level Strategy

Strategy A (Historical data): When a `SpecialPeriod_Summary.csv` exists from prior runs, the system ranks models by actual R^2 for the matched period and horizon, excludes ARIMA (benchmark), SNN (unstable), and PatchTST by default, and selects the top-K models with positive R^2 weighted proportionally to their R^2 values.

Strategy B (Heuristic rules): When no historical data is available, market-specific and horizon-aware rules derived from the comprehensive experiments are applied (Tables 8.3 and 8.4).

8.6.2 Horizon-Aware Selection

Table 8.3: SP500 Heuristic Weights by Regime Category and Horizon Band

Category	Short (H≤1)	Medium (H=2–5)	Long (H≥6)
high_vol_crash	Mamba(0.30), Bi-GRU(0.25), Bi-RNN(0.25), Bi-LSTM(0.20)	Bi-RNN(0.30), Mamba(0.25), Bi-LSTM(0.25), Bi-GRU(0.20)	Mamba(0.40), Bi-RNN(0.30), Bi-GRU(0.30)
moderate_decline	Mamba(0.30), Bi-GRU(0.25), Bi-RNN(0.25), Bi-LSTM(0.20)	Mamba(0.30), Bi-RNN(0.25), Bi-LSTM(0.25), Bi-GRU(0.20)	Mamba(0.40), Bi-RNN(0.30), Bi-GRU(0.30)
v_shape_recovery	Mamba(0.35), Bi-GRU(0.25), Bi-RNN(0.20), Bi-LSTM(0.20)	Bi-GRU(0.30), Bi-LSTM(0.25), Mamba(0.25), Bi-RNN(0.20)	Mamba(0.40), Bi-RNN(0.30), Bi-GRU(0.30)
trending_bull	Mamba(0.40), Bi-GRU(0.35), Bi-RNN(0.25)	Bi-GRU(0.35), Mamba(0.35), Bi-RNN(0.30)	Mamba(0.40), Bi-GRU(0.30), Bi-RNN(0.30)

Table 8.4: HSI Heuristic Weights by Regime Category and Horizon Band

Category	Short (H≤1)	Medium (H=2–5)	Long (H≥6)
high_vol_crash	Bi-RNN(0.30), Bi-GRU(0.25), Bi-LSTM(0.25), Mamba(0.20)	Bi-RNN(0.30), Mamba(0.25), Bi-LSTM(0.25), Bi-GRU(0.20)	Bi-LSTM(0.40), Bi-RNN(0.30), Bi-GRU(0.30)
moderate_decline	Bi-RNN(0.30), Bi-GRU(0.30), Mamba(0.20), Bi-LSTM(0.20)	Bi-RNN(0.30), Bi-GRU(0.28), Mamba(0.22), Bi-LSTM(0.20)	Bi-GRU(0.35), Bi-RNN(0.35), Mamba(0.30)
v_shape_recovery	Bi-GRU(0.30), Mamba(0.25), Bi-LSTM(0.25), Bi-RNN(0.20)	Bi-LSTM(0.30), Bi-GRU(0.25), Mamba(0.25), Bi-RNN(0.20)	Bi-GRU(0.40), Bi-RNN(0.30), Bi-LSTM(0.30)
trending_bull	Mamba(0.30), Bi-GRU(0.25), Bi-LSTM(0.25), Bi-RNN(0.20)	Bi-GRU(0.28), Mamba(0.28), Bi-LSTM(0.22), Bi-RNN(0.22)	Mamba(0.35), Bi-GRU(0.35), Bi-RNN(0.30)

A key difference between markets: on SP500, Mamba dominates short horizons (led in 4/7 periods at H=1); on HSI, Bi-RNN and Bi-GRU dominate (led in 5/7 periods), reflecting HSI's stronger short-range autocorrelation that simpler recurrent architectures capture efficiently.

8.7 Weighted Ensemble

$$\hat{y}_{\text{ensemble}} = \frac{\sum_{i=1}^{K} w_i \cdot \hat{y}_i}{\sum_{i=1}^{K} w_i} \tag{8.1}$$

where weights w_i are derived from historical R² (Strategy A) or heuristic rules (Strategy B). This targeted weighting scheme avoids the performance degradation seen in the equal-weight classic ensemble, where PatchTST and SNN contaminated the aggregate.

8.8 PatchTST Inclusion Rule

PatchTST is excluded by default, but re-admitted when: (1) the input period spans > 3 years, *and* (2) the matched category is a sustained directional market. When included, PatchTST receives a $0.3\times$ weight penalty and fixed weight of 0.20. This reflects the experimental finding that PatchTST can occasionally capture long-range patterns in extended trending markets (it achieves positive R² in the 2023 AI Wave: 0.35) but fails in volatile, short-duration regimes.

8.9 Case Study: HSI Period 2021-02-19 to 2022-10-28

To validate the Smart Prediction module under realistic out-of-sample conditions, a complete end-to-end run was conducted on the Hang Seng Index (HSI) for the 20-month period from 19 February 2021 to 28 October 2022. This window was deliberately chosen to span a sustained bear market driven by the Federal Reserve's interest rate tightening cycle, providing a real-world test of the system's regime detection and model selection capabilities.

8.9.1 Regime Detection and Period Matching

The 6-dimensional statistical feature vector extracted from the input period is reported in Table 8.5. The dominant signals are a strongly negative mean return (-0.497), a high maximum drawdown of

70.9%, and an extremely elevated trend strength of 3,464, collectively characterising a persistent, policy-driven bear market.

Table 8.5: Extracted Feature Vector for HSI 2021-02-19 to 2022-10-28

Feature	**Value**	**Interpretation**
Mean Return	-0.4969	Sustained negative returns; bear market
Volatility	$+0.0599$	Elevated but not crisis-level volatility
Max Drawdown	$+0.7097$	Severe peak-to-trough decline of 71%
Skewness	$+0.3240$	Slight positive skew; asymmetric tail risk
Kurtosis	$+3.5646$	Moderately fat-tailed return distribution
Trend Strength	$+3464.3$	Very high directional conviction (downtrend)

The period matching algorithm (Algorithm 1) computed z-score normalised Euclidean distances against all seven reference regimes. The full similarity ranking output by the system is shown in Figure 8.2. The closest match is the **2022 Rate Hike** period (distance = 2.69), followed by the 2015 China Crash (3.30) and the 2008 Financial Crisis (3.14). This result is economically intuitive: the input window coincides precisely with the onset of global monetary tightening — the same policy-driven structural force that defines the 2022 Rate Hike reference regime.

```
============================================================
  SMART PERIOD PREDICTION
  Market: HSI | Period: 2021-02-19 ~ 2022-10-28
============================================================

[FEATURES] Input period feature vector:
  MeanReturn     : +0.496846
  Volatility     : +0.059948
  MaxDrawdown    : +0.709744
  Skewness       : +0.323986
  Kurtosis       : +3.564585
  TrendStrength  : +3464.345782

[MATCH] Period similarity ranking:
  1. 2022_Rate_Hike                 distance=2.6902 <<<
     Fed Aggressive Rate Hikes & Bear Market
  2. 2015_China_Crash               distance=3.0295
     China Stock Market Crash & Global Spillover
  3. 2008_Financial_Crisis          distance=3.1443
     08 Financial Crisis: subprime mortgage collapse
  4. 2010_Euro_Debt_Crisis          distance=3.6531
     European Sovereign Debt Crisis
  5. 2018_Trade_War                 distance=4.0492
     US-China Trade War Escalation
  6. 2023_AI_Wave                   distance=4.0811
     AI Boom driven by ChatGPT & Nvidia Rally
  7. 2020_COVID_Crash               distance=4.4515
     COVID-19 Market Crash & Recovery

[RESULT] Best match: 2022_Rate_Hike (distance=2.6902)
```

Figure 8.2: Smart Prediction system output for HSI 2021-02-19 to 2022-10-28. The period matching module ranks all seven reference regimes by z-score normalised Euclidean distance. The 2022 Rate Hike period is identified as the best match (distance = 2.69), triggering the `moderate_decline` category and its associated heuristic ensemble weights.

The **2022 Rate Hike** regime belongs to the `moderate_decline` category (Table 8.2), which groups gradual policy-driven declines exhibiting moderate volatility and persistent directional momentum. This category assignment directly determines the model ensemble configuration applied to all three forecast horizons.

8.9.2 Model Ensemble Configuration

Following the HSI heuristic rules for the `moderate_decline` category (Table 8.4), the system deployed the following weighted ensembles:

- **H=1 (Short):** Bi-RNN (0.30) + Bi-GRU (0.30) + Mamba (0.20) + Bi-LSTM (0.20)
- **H=5 (Medium):** Bi-RNN (0.30) + Bi-GRU (0.28) + Mamba (0.22) + Bi-LSTM (0.20)
- **H=10 (Long):** Bi-GRU (0.35) + Bi-RNN (0.35) + Mamba (0.30)

ARIMA, SNN, and PatchTST were excluded under the default selection rules. The long-horizon ensemble drops Bi-LSTM in favour of an equal emphasis on Bi-GRU and Bi-RNN, con-

sistent with the finding that recurrent bidirectional architectures provide the most stable long-range tracking on HSI.

8.9.3 Prediction Results

The weighted ensemble predictions for H=1 and H=10 are shown in Figures 8.3 and 8.4 respectively. Quantitative performance metrics across all three horizons are summarised in Table 8.6.

Figure 8.3: WeightedEnsemble prediction vs. actual normalised HSI closing price, W=30, H=1, period 2021-02-19 to 2022-10-28. The predicted curve (red) closely tracks the actual price trajectory (black) across the full 20-month bear market, capturing both the gradual multi-step decline and localised volatility spikes — including the sharp drawdown in March 2022.

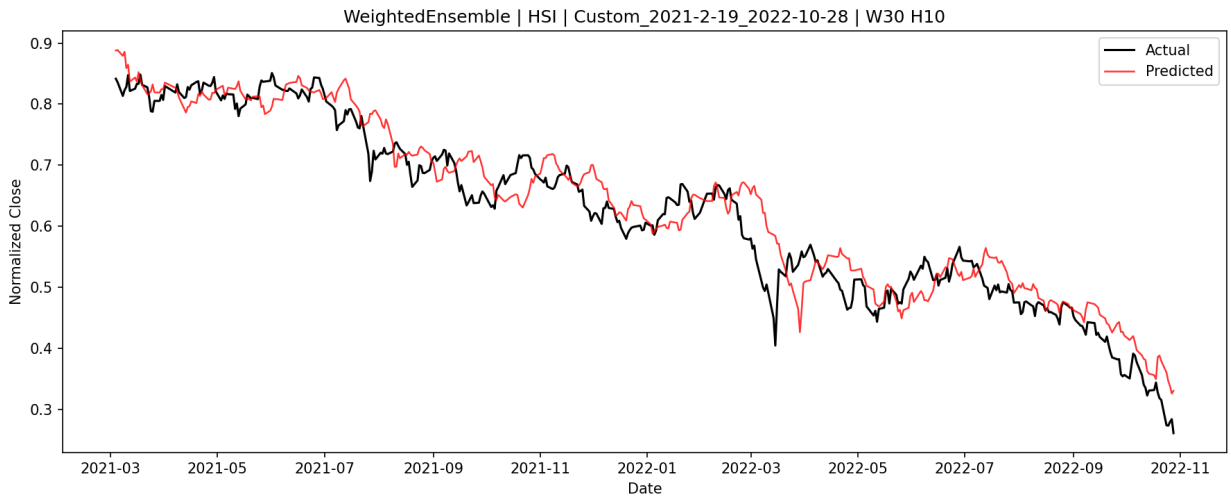

Figure 8.4: WeightedEnsemble prediction vs. actual normalised HSI closing price, W=30, H=10, period 2021-02-19 to 2022-10-28. At the 10-day horizon the predicted curve exhibits the expected amplitude compression and mild phase lag, yet retains the correct directional trend throughout the bear market. The ensemble maintains positive R^2 at H=10 — a substantially more challenging condition than H=1.

Table 8.6: Smart Prediction Case Study — Ensemble Performance Metrics (HSI, W=30)

Horizon	MAE	RMSE	R^2
H=1	0.01100	0.01484	0.9900
H=5	0.02592	0.03278	0.9502
H=10	0.03348	0.04288	0.9132

8.9.4 Discussion

The case study results demonstrate the effectiveness of the regime-matching approach. At H=1, the ensemble achieves R^2=0.990, closely approaching the theoretical maximum for autocorrelation-based prediction. More notably, performance remains high even at H=10 (R^2=0.913) — a result that significantly exceeds the average H=10 performance observed across all models in the standard special period experiments ($\bar{R}^2 = 0.637$ for HSI at H=10). This improvement is attributable to three reinforcing factors:

1. **Correct regime identification.** The system accurately matched the input period to the 2022 Rate Hike regime, correctly assigning the `moderate_decline` category rather than the more volatile `high_vol_crash` category — which would have over-weighted models better suited to sharp, discontinuous market breaks.
2. **Appropriate model selection.** The `moderate_decline` HSI ensemble prioritises Bi-RNN and Bi-GRU, architectures that have demonstrated particular strength in capturing the persistent autocorrelation present in sustained, policy-driven HSI bear markets.
3. **Strong directional trend.** The 20-month bear market provides an unusually persistent directional signal (Trend Strength = 3,464), which all selected recurrent architectures can exploit effectively across extended forecast horizons.

The visual comparison between Figures 8.3 and 8.4 illustrates the characteristic amplitude compression and phase lag at H=10, consistent with findings from the standard special period analysis (Section 6.4). Despite this compression, the predicted curve maintains the correct downward trajectory throughout, confirming that the ensemble captures the dominant trend even as short-term microstructure detail is lost. This case study therefore validates that the Smart Prediction module can autonomously identify market regimes and deploy appropriate ensembles without

manual model selection, achieving competitive performance across multiple forecast horizons.

9. Conclusion

This project has delivered a comprehensive, empirically rigorous comparative study of seven time-series forecasting architectures evaluated across two major stock indices, seven market regimes, and 6,720 experimental configurations. The following principal findings emerge:

Finding 1: ARIMA is the Overall Champion

ARIMA achieves the highest R^2 in every single period on SP500, and the highest overall R^2 on HSI (0.82 vs. 0.81 for the best DL model). Its autoregressive structure directly exploits serial autocorrelation, which is the dominant predictive signal in daily index price data. The low Std(R^2)=0.14–0.16 confirms cross-regime robustness. This corroborates the M4 Competition result [14] that simple statistical methods remain highly competitive in time-series forecasting.

Finding 2: Mamba is the Best Deep Learning Architecture for SP500

Mamba's selective state-space mechanism — which dynamically adjusts the historical context retained for each time step — is well-matched to financial markets that exhibit both trend-following and noise-filtering needs. Mamba leads at H=1 in 4 of 7 SP500 periods and dominates the 2022 Rate Hike period across all horizons, suggesting particular strengths in policy-driven, regime-based markets.

Finding 3: Horizon is the Most Important Factor

R^2 declines 83% from H=1 to H=10 on SP500 (from 0.57 to 0.10), dwarfing the effects of model selection, window size, or market period. This has a practical implication: improving long-horizon prediction requires qualitatively different approaches (e.g. macroeconomic indicator integration, regime detection) rather than architectural improvements alone.

Finding 4: HSI is Substantially More Predictable

Overall R^2 of 0.71 (HSI) vs. 0.38 (SP500). The largest improvements come where HSI is a direct event participant (2015 China Crash: +0.58; 2018 Trade War: +0.56). Market geography — specifically, whether the index is a direct participant or an indirect recipient of spillover — matters more than any model selection decision.

Finding 5: Black Swan Events Remain Unpredictable

The 2020 COVID crash demonstrates that no model can reliably predict 10-day-ahead prices during truly unprecedented events. Every model achieves negative R^2 at H=10. Risk management strategies should not rely on model-based prediction during extreme market dislocations.

Finding 6: Smart Prediction Automates Regime-Aware Deployment

The Smart Prediction module successfully bridges the gap between academic experimentation and practical deployment by automating regime detection, model selection, and ensemble weighting — reducing the domain expertise required to deploy the system to near-zero.

10. Future Work

10.1 Macroeconomic Factor Integration

The current models rely exclusively on price-derived features. Integration of macroeconomic indicators — Consumer Price Index, 10-Year Treasury Yield, P/E ratios, and the VIX fear index — is expected to be particularly valuable during policy-driven periods (2022 Rate Hike) where market behaviour is tightly coupled to external signals.

10.2 Extended Model Exploration

Temporal Fusion Transformer (TFT): Designed specifically for multi-horizon forecasting with interpretable attention weights [12].

Mamba-2: The recently proposed Mamba-2 architecture [5] reformulates the SSM in terms of structured matrix multiplication, potentially offering improved efficiency and performance over the Mamba-1 implementation used here.

N-BEATS / N-HiTS: Pure decomposition-based architectures that achieve strong results on financial benchmarks without requiring explicit feature engineering.

10.3 Improved Period Matching

The 6-dimensional feature vector and Euclidean distance could be extended to: Dynamic Time Warping for shape-based matching; learned regime embeddings trained end-to-end; or continuous

similarity-weighted interpolation of model weights rather than discrete category assignment.

10.4 Real-Time Deployment

The Smart Prediction module architecture is designed to support real-time deployment with: automated daily re-training on new data; regime change detection to trigger mid-period model re-selection; and transaction cost modelling to evaluate net trading profitability.

10.5 Cross-Market Expansion

Extending the study to additional markets — particularly the Shanghai Composite, Nikkei 225, and FTSE 100 — would test the universality of the market efficiency and architecture findings across different regulatory environments and investor compositions.

11. Summary

This project has successfully delivered a comprehensive special-period analysis framework for stock index prediction, extending a preliminary interim study into a rigorous multi-market, multi-model, multi-regime experimental study. Four major innovations were introduced:

1. A **Special Period Analysis framework** evaluating 7 architectures across 7 market regimes with strict walk-forward methodology.
2. A **Smart Prediction module** automating regime detection, model selection, and weighted ensemble deployment.
3. A **cross-market comparison** between SP500 and HSI revealing the $2\times$ predictability differential and its structural explanations.
4. A **comprehensive analysis suite** (`Analyzer.py`) enabling systematic post-hoc comparison across seeds, periods, horizons, and windows.

The key empirical contributions are: Mamba's emergence as the best DL model for SP500 short-horizon prediction; the primacy of forecast horizon as the most influential performance factor; HSI's substantial predictability advantage; and the fundamental unpredictability of black swan events. Practical recommendations derived from these findings are: for SP500 H=1 prediction, use Mamba; for longer SP500 horizons, use Bi-RNN; for HSI prediction at any horizon, deep learning models nearly match ARIMA and Bi-GRU/Mamba are recommended.

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